

SMEsPlus SaaS Development Readiness Blueprint v0.2

1. System Architecture 11-Layer

1. UI Layer: Web app, mobile-responsive, role-based navigation
2. API Gateway: Auth, rate limit, request validation, routing
3. Application Service: Orchestration, DTO validation, transaction boundary
4. Business Domain: Accounting, Sales, Inventory, HR, CRM
5. Workflow & Approval: Approval rules, task queue, escalation
6. Integration Layer: REST API, webhook, Make.com, external services
7. Data Access Layer: Repository pattern, tenant filtering
8. Database Layer: PostgreSQL, schema-per-tenant, RLS, audit trail
9. Object Storage: Tenant-aware document storage
10. Infrastructure: Docker, Kubernetes, Redis, queue, MinIO/S3
11. Observability: Logs, metrics, tracing, audit, alerting

2. MVP Module Boundary

MVP ควรเริ่มจาก Foundation Suite ก่อน:

- Tenant Management
- Company / Branch Management
- Identity & Access Management
- Subscription & Package Management
- Module Activation
- Configuration Center
- Approval Foundation
- Notification Foundation
- Audit & Governance
- Integration Foundation

Phase 2 จึงเพิ่ม business modules:

- Accounting
- Sales
- Inventory
- Purchasing
- HR
- CRM
- Reporting / BI

3. Core ERD Concept

Core entities:

- tenants
- companies

- branches
- divisions
- users
- roles
- permissions
- user_role_assignments
- subscriptions
- packages
- modules
- tenant_module_activations
- approval_workflows
- notifications
- audit_logs
- api_clients
- integration_events

ทุก business table ต้องมี:

- tenant_id
- company_id
- branch_id ถ้าเกี่ยวข้อง
- created_at
- created_by
- updated_at
- updated_by
- deleted_at
- version

4. API Contract Baseline

มาตรฐาน endpoint:

- GET /api/v1/tenants
- POST /api/v1/tenants
- GET /api/v1/companies
- POST /api/v1/companies
- GET /api/v1/users
- POST /api/v1/users
- GET /api/v1/modules
- POST /api/v1/tenant-modules/activate
- GET /api/v1/audit-logs
- POST /api/v1/webhooks/register

มาตรฐาน response:

```
{
  "success": true,
  "data": {},
  "meta": {
    "request_id": "uuid",
  }
}
```

```

    "tenant_id": "tenant_x",
    "timestamp": "ISO-8601"
  },
  "errors": []
}

```

5. Permission Matrix

Role	Scope	Key Access
Platform Admin	Platform	manage tenants, packages, billing
Tenant Owner	Tenant	manage companies, users, modules
Company Admin	Company	manage branches, users, reports
Branch Manager	Branch	branch operation data
Department User	Division	assigned module actions
Auditor	Tenant/Company	read-only audit and reports

กฎสำคัญ: ห้ามใช้ frontend เป็นตัวบังคับ permission หลัก ต้อง enforce ที่ backend และ database RLS

6. Key Screens

Dashboard

目的: ให้ผู้ใช้เห็นสถานะรวมของ tenant/company/branch

機能: KPI cards, pending approvals, notifications, recent activities, module shortcuts

Tenant Administration

目的: ให้ platform admin จัดการ tenant

機能: create tenant, suspend tenant, plan assignment, usage quota, tenant health

Company / Branch Management

目的: จัดการ legal entity และสาขา

機能: company profile, tax ID, branch code, address, reporting hierarchy

User & Role Management

目的: จัดการผู้ใช้และสิทธิ์

機能: invite user, assign role, MFA, deactivate user, permission audit

Module Activation

目的: เปิด/ปิด module ตาม package

機能: module catalog, activation status, dependency check, billing impact

Configuration Center

目的: ตั้งค่าระบบราย tenant/company

機能: tax settings, fiscal year, numbering series, approval rules, localization

Approval Inbox

目的: รวมงานรออนุมัติ

機能: approve/reject, comment, escalation, delegation

Audit & Governance

目的: ตรวจสอบการใช้งานและ compliance

機能: audit log, filter by user/action/module, export, anomaly flag

Integration Center

目的: จัดการ API/webhook/external tools

機能: API key, webhook registration, event log, retry status

Reports

目的: ดูรายงานและ export

機能: filter, saved report, PDF/Excel export, scheduled delivery

7. Non-Functional Requirements

- Availability: 99.9%
- RTO: < 1 hour
- RPO: < 15 minutes
- Page load: < 1.2 seconds
- List query: < 300 ms
- Create record: < 500 ms
- API throughput: 500+ req/sec target
- Tenant boundary violation = critical defect
- All critical operations must produce audit logs

8. DevOps Specification

Environments:

- local
- dev
- staging
- production

CI/CD gates:

1. lint
2. unit test

3. integration test
4. tenant isolation test
5. security scan
6. migration dry run
7. staging deployment
8. approval gate
9. blue-green production deployment
10. rollback verification

9. Test Strategy

Required test suites:

- Unit test: business logic
- API test: endpoint contract
- Integration test: service-to-service
- Security test: auth, permission, tenant isolation
- Performance test: load, stress, p95/p99
- Migration test: schema upgrade/rollback
- UAT test: role-based business flow
- DR test: backup restore and failover

10. ADR Required Before Development

ต้องอนุมัติ ADR เหล่านี้ก่อนเริ่ม implementation:

- ADR-0010 SaaS First
- ADR-0011 Multi-Tenant by Design
- ADR-0012 API First
- ADR-0013 Schema-per-Tenant Strategy
- ADR-0014 RBAC + ABAC + RLS Authorization
- ADR-0015 Event/Webhook Integration
- ADR-0016 Deployment via Kubernetes
- ADR-0017 Audit by Design
- ADR-0018 Thai Localization Foundation

Development Readiness Verdict

สถานะ: READY FOR SDS / API / DB / UI SPEC

ยังไม่ควรเริ่ม coding ทั้งระบบทันที แต่สามารถเริ่มทำรายละเอียดระดับต่อไปได้:

1. SDS: Software Design Specification
2. OpenAPI Spec
3. ERD ราย module
4. UI wireframe
5. Migration script baseline
6. Test case repository
7. Sprint backlog สำหรับ Foundation MVP